

1 Memory Subsystem

The AM62x memory subsystem integrates SRAM, ECC, and cache controllers to provide high-reliability data storage for safety-critical applications. All memory regions support ECC protection by default.

This chapter covers ECC configuration registers, SRAM controller setup, and the recommended initialization sequence.

1.1 ECC Configuration

Error Correction Code (ECC) provides single-bit error correction and double-bit error detection (SECEDED) for all SRAM regions. ECC is enabled via the ECC_CTRL register at base address 0x40000000.

To enable ECC: set bit 0 (ECC_EN) of ECC_CTRL. On detection of a correctable error the ECC_STATUS register bit 0 is set and an interrupt is raised if ECC_IRQ_EN is configured. The faulting address is latched in ECC_ERR_ADDR. Software must clear ECC_STATUS before re-enabling.

Register map: ECC_CTRL = 0x40000000 (RW) ECC control ECC_STATUS = 0x40000004 (RO) Error status ECC_ERR_ADDR = 0x40000008 (RO) Faulting address ECC_IRQ_EN = 0x4000000C (RW) Interrupt enable

1.2 SRAM Controller

The SRAM controller manages access arbitration between CPU, DMA, and peripheral masters. Up to 4 banks of 256 KB each are available. Banks can be powered down independently for low-power operation.

Initialization sequence: (1) Configure SRAM_CFG base address and size. (2) Enable ECC if required. (3) Zero-fill memory to initialize ECC syndrome bits. (4) Enable SRAM_CTRL.EN. The driver sram_init() implements this sequence.

2 DMA Controller

The Direct Memory Access (DMA) controller provides eight independent channels capable of memory-to-memory, memory-to-peripheral, and peripheral-to-memory transfers. Each channel has a 16-entry descriptor FIFO for chained operations.

2.1 DMA Channel Configuration

Each DMA channel is configured via a channel descriptor structure: source address, destination address, transfer count, and control word. The control word specifies transfer width (8/16/32-bit), burst length, and address increment mode.

To start a transfer: (1) Write descriptor to DMA_DESC[ch]. (2) Set DMA_CTRL[ch].EN. (3) Poll DMA_STATUS[ch].DONE or enable DMA_IRQ_EN[ch] for interrupt-driven operation.

2.2 DMA Transfer Modes

Single-shot mode: one descriptor, channel halts after completion. Circular mode:

channel automatically reloads the same descriptor, useful for ADC sampling.
Chain mode: DMA_DESC.NEXT_DESC points to the next descriptor, enabling scatter-gather without CPU intervention.

All modes support 32-bit, 16-bit, and 8-bit element widths. Misaligned transfers are not supported and will raise a bus error logged in DMA_ERR_STATUS.